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Always Room, Except When There Isn’t

Auditing a Share-House Parcel-Locker Bank’s Hold Guarantee Under Load

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Background

Share-house parcel-locker banks promise a slot for as long as a resident needs it. Despite growing reliance on shared locker infrastructure, little is known about what “held” means once a bank nears capacity — does a reserved slot really wait, or does it just look like it does?

Objectives

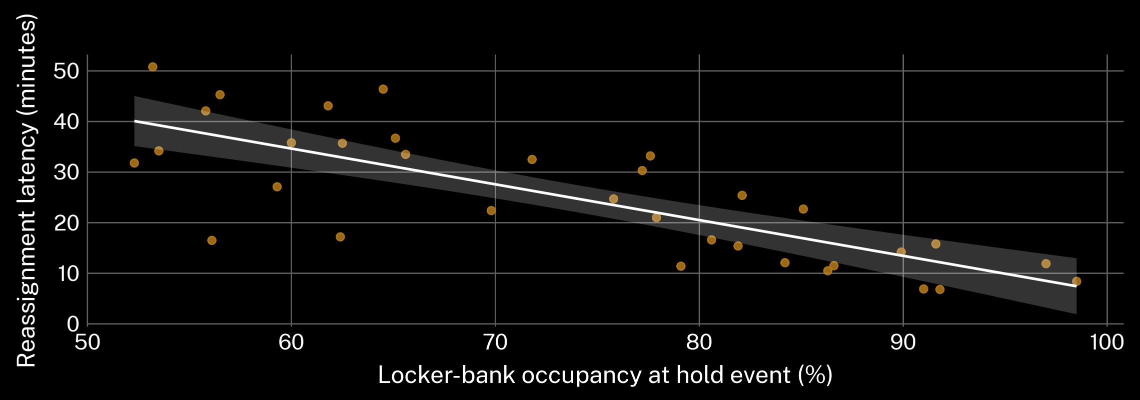
- Quantify slot-reassignment latency as locker-bank occupancy approaches capacity.
- Compare the app’s displayed hold status against the backend reassignment timestamp.
- Locate the occupancy threshold at which “held” stops meaning held.

Materials & methods

- n = 6 share houses (9-16 residents each) · locker-event logs · 5-min polling · 16-week deployment
- hold events matched to reassignment timestamps by slot ID; app-displayed status logged independently of backend state
- occupancy threshold pre-registered at 85%; no post hoc adjustment

Results

Reassignment latency held roughly steady below 70% occupancy, then fell sharply as the bank filled. The in-app hold badge continued to read “reserved” for a median of 6 minutes after the backend had already reassigned the slot.



Median reassignment latency fell from 41 minutes at 60% occupancy to 6 minutes at 97% occupancy across 1,842 hold events logged March-June 2026 (n = 6 share houses); the in-app badge updated a median of 6 minutes after the backend reassignment, rising to 11 minutes above 90% occupancy.

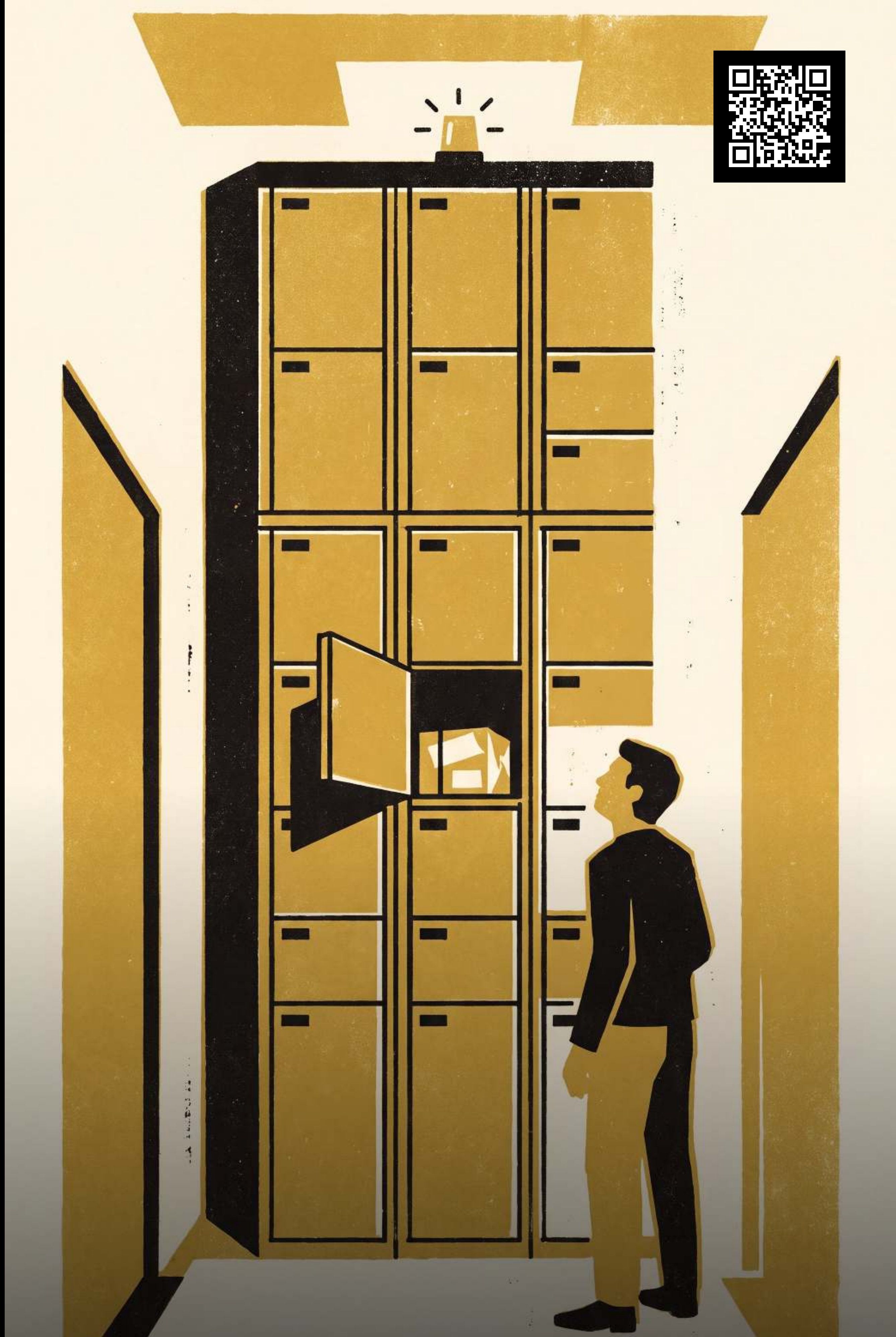
Discussion

Findings are consistent with a hold guarantee that degrades under load rather than one that holds absolutely. The badge lag looks structural, not incidental: reassignment fires before the app is told to update. Next: instrument the notification queue directly to separate network delay from policy delay.

Further reading

1. Mancini, S., & Gansterer, M. (2026). Dynamic stochastic parcel locker assignment with uncertain pick-up times. Omega, 140, 103478. doi:10.1016/j.omega.2025.103478
2. Shajin, D., & Krishnamoorthy, A. (2021). On a queueing-inventory system with impatient customers, advanced reservation, cancellation, overbooking and common life time. Operational Research, 21(2), 1229–1253. doi:10.1007/s12351-019-00475-3
3. Birkenheuer, G., & Brinkmann, A. (2011). Reservation-based overbooking for HPC clusters. 2011 IEEE International Conference on Cluster Computing, 537–541. doi:10.1109/cluster.2011.74
4. Jin, Z., Tao, D., Qi, P., & Gao, R. (2024). An adaptive cloud resource quota scheme based on dynamic portraits and task-resource matching. IEEE Transactions on Cloud Computing, 12(4), 996–1010. doi:10.1109/tcc.2024.3410390
5. Beng, C., Hanke, M., & Nwosu, J. (2026). Notified, Not Yet Loaded: Predictive Ready-for-Pickup Alerts Against a Parcel Locker’s Own Door-Sensor Confirmation. School of Continuous Improvement. doi:10.5555/slop.yqm3vd

De-identified slot-event logs only; no resident names or unit numbers retained; thresholds pre-registered.



“The app still says ‘held.’”

Median reassignment latency fell to 6 minutes once the bank passed 90% occupancy (n = 6 share houses, 16-week deployment).